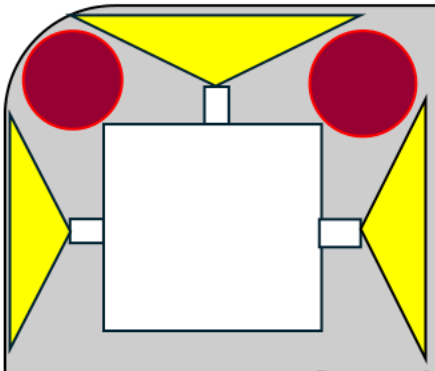
Poster for robot car

Group members: Shane Whittaker, Eli Dick,

Shawn Wang and Surjo Datta

Introduction and objective:

The aim of this project was to build an autonomous robot car that could navigate a maze using only circuitry. The car needed sensors to detect obstacles and a logic unit to process the inputs from the sensor and connect to the motor drive to determine the direction the car would move.



Big square = car body

Small rectangles = sensors

triangle shapes = area of visibility of sensors

Red circles = blind spots

Pros and cons of our design:

Pros:

- This design keeps everything simple.
- 3 sensors cover most of the car

cons:

- Small blind spots on the front corners of the car

Components used:

sfh 203 fa (IR photodiode ±20 degree cone)

sfh 4545 (IR LED ±30 degree cone)

Lm293 (h-bridge)

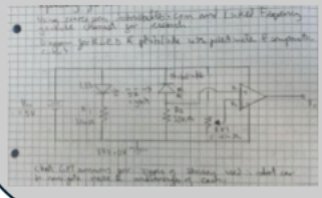
Lm358n (IC circuit)

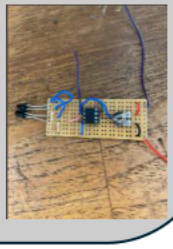
Sensor for car:

Our sensors pick up a signal if the wall is 4.7-5.1cm away from it

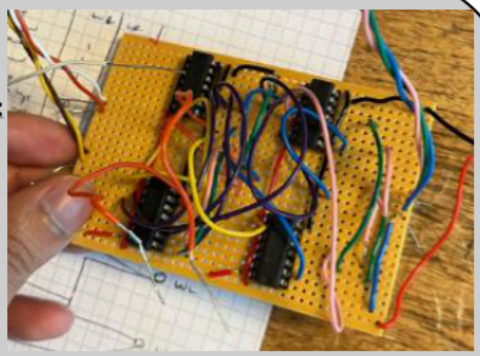
The three sensors we use cover almost all the cars front, minor blind spots.

IR LEDs and photodiodes chosen to give the car a faster response time





Logic circuit for car:



Design for the logic circuit:

Inputs for the circuit are if any of the sensors detect a wall or not

Wall detected = 1

wall not detected = 0

F = Forward

L = Left

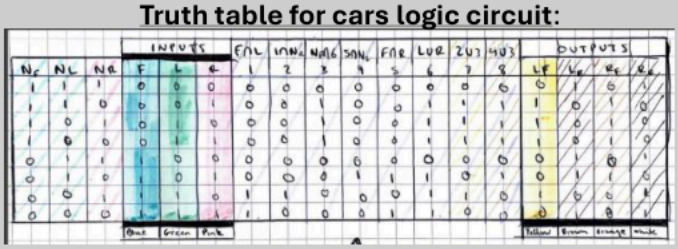
R = Right

NF = Not Forward

NL = Not Left

NR = Not Right

Truth table for cars logic circuit:



Challenges faced:

- Initial design of h-bridge did not work, after soldering step. Could be due to soldering over lapping with each other.
- Experiments/ time off often interrupted our lab sessions, making it harder to remember what was needed to be done next.
- Connection of wires where designed in a complicated way, making it difficult to understand other group members circuits.
- Team was not experienced in group projects like this
- Broken chips

Motor and h-bridge:

Lm293 has voltage range in that our 6V falls into.

It is a dual bridge meaning that it can be to control two motors, perfect for our car.

Has built in protection, to protect from inductive loads like motors.

